The Role of Tension Control in Gravure Printing: Historical Evolution, Current Methods and Guideline Tension Values from Unwind to Rewind

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***Abstract*—In rotogravure printing, register accuracy, print quality and roll quality all depend on holding web tension steady in every zone from unwind to rewind. This paper traces how tension control evolved: from hand-set friction brakes and mechanical line shafts, through dancers, powder brakes and analog loops, to today's electronic-line-shaft presses with servo drives, load cells and digital multi-zone control. It then gives practical guideline tension values for common gravure substrates (BOPET, BOPP, CPP, polyethylene, nylon, shrink films, aluminium foil, paper and board), scaled by thickness and web width, with zone-wise ratios from unwind through infeed, print units and outfeed to rewind taper. The values are starting points to be refined on each press against register, wrinkle and roll-quality evidence.**

***Index Terms*—Rotogravure, web tension, tension zones, dancer, load cell, electronic line shaft, taper tension, register control, flexible packaging.**

I. Introduction

TENSION is the one process variable that touches every other variable on a gravure press. A rotogravure press prints from engraved cylinders, one colour per unit, with a dryer and chill roll after each unit. Because of the dryer, the web travels several metres between successive impression nips. Over that path, any change in tension changes the web's elastic stretch, and so the image printed by one unit lands early or late against the next.

The press is built as a chain of units—unwind, infeed, print units, outfeed and rewind—each driven or braked separately [1]. Each driven point bounds a tension zone, and the controller in each zone must reject its own disturbances without passing them along.

The stakes are practical:

- *Register:* gravure work is commonly held to about ± 0.1 mm between colours; a tension swing of a few newtons on a stretchy film moves the image by more than that (Section II).
- *Print defects:* low tension lets the web flutter, wrinkle and pick up ink out of place; high tension stretches thermoplastic films in hot dryers and causes neck-in, curl and repeat-length error.
- *Roll quality and downstream yield:* wrong rewind tension produces telescoping, starring, blocking, or gauge bands that spoil lamination and pouch making.
- *Waste:* most register waste is made during start-up, splicing, acceleration and deceleration—exactly the transients the tension system must ride through.

TABLE I
Register Shift from a 10 N Tension Step (1000 mm Web, 4 m Span)

Web	hW (mm^2)	$\Delta\sigma$ (MPa)	E (GPa)	$\Delta\varepsilon$	Shift
BOPET 12 µm	12	0.83	4.5	1.9×10^{-4}	≈0.7 mm
LDPE 50 µm	50	0.20	0.2	1.0×10^{-3}	≈4 mm

This paper sets out the physics briefly (Section II), follows the historical evolution of the hardware (Section III), describes current practice (Section IV) and the tension zones of a press (Section V), gives guideline values (Section VI), and closes with commissioning and troubleshooting guidance (Section VII).

II. Fundamentals of Web Tension

Web tension is the machine-direction force carried by the web; what damages or distorts the web is the stress and strain it produces. For a web of thickness h and width W running at tension T,

$$\sigma = \frac{T}{h\,W}, \qquad \varepsilon = \frac{\sigma}{E} \tag{1}$$

where σ is the stress, ε the elastic strain and E the machine-direction modulus. Three consequences follow.

1) *Tension scales with width and thickness.* Guideline values are therefore quoted per unit width (N/m, kgf/cm, or PLI in US practice) and, for films, per unit thickness. A useful identity: 1 N/m per µm of thickness is exactly 1 N/mm^2 (1 MPa) of stress.
2) *Stretchy webs are tension-sensitive.* The same tension change produces a strain change inversely proportional to E. Polyethylene ($E \approx 0.2$ GPa) strains about twenty times more than polyester ($E \approx 4$–5 GPa) for the same stress.
3) *Strain becomes register error.* If tension between two print units changes, the web length held between them changes by $L\,\Delta\varepsilon$, and the second colour moves by roughly that amount until the zone settles.

Table I illustrates a 10 N tension step on a 1000 mm wide web with $L = 4$ m of web between units. Both shifts exceed a ± 0.1 mm register tolerance, which is why register and tension controllers must work together, and why polyethylene is the hardest common gravure substrate to hold in register.

TABLE II
UNITS AND CONVERSIONS USED IN THIS PAPER

Quantity	Conversion
Tension per width	1 PLI (lbf/in) = 175.1 N/m; 1 kgf/cm = 981 N/m
Specific tension (films)	1 PLI/mil = 6.895 N/m per µm = 6.895 MPa
Total tension	T (N) = specific tension (N/m per µm) $\times$ h (µm) $\times$ W (m); 1 kgf = 9.81 N
Shaft torque	M (N·m) = T (N) $\times$ D (m)/2

A. Tension Zones and Disturbances

A tension zone is the web span between two points that set web speed or force: a braked or driven roll, or a nip. On a gravure press these are the unwind shaft, the infeed draw rolls, each impression nip, the outfeed draw rolls and the rewind shaft. Tension inside a zone is set by the speed difference across it (draw) and by the torque applied at its ends. The main disturbances are:

- *Changing roll diameter:* unwind brake torque must fall as the roll shrinks and rewind torque must rise as it grows. At 100 N, unwind torque falls from about 20 N·m at a 400 mm roll to about 5 N·m at 100 mm [2].
- *Roll eccentricity and out-of-round cores:* a once-per-revolution tension ripple.
- *Acceleration, deceleration and flying splices:* inertia of rolls and dancers adds transient torque.
- *Heat and moisture:* thermoplastic films soften in dryers (typically 40–120 °C), and paper changes length with moisture [2].
- *Changing cylinder diameters:* each job's engraved cylinders differ slightly in circumference, which changes the draw between units.
- *In-line lamination* of extra plies, which adds tension to the combined web [9].

III. HISTORICAL EVOLUTION OF TENSION CONTROL

Tension control moved from the operator's hand to closed-loop digital control in four broad stages, each driven by faster presses and thinner, stretchier substrates (Table III). The eras are approximate; adoption varied widely by country and press maker.

A. The Mechanical Era

On early presses a single main motor drove every printing cylinder through a line shaft and bevel gears, so the units were synchronised mechanically. The unwind was held back by a simple brake—a leather band on a flywheel is a commonly cited example—and early unwinding brakes were adjusted by hand from the operator's feel [2]. Because brake torque was roughly fixed, tension rose as the unwind roll shrank.

The rewind was driven from the main drive through a slip clutch, overspeeding the web so the clutch always slipped. A fixed clutch torque gives tension that falls in inverse proportion to roll diameter. Patent literature on conventional presses notes that unwind braking and rewind advance were not governed by the main drive shaft, so that mid-roll the press became starved at the feed end and under-tensioned at the rewind [12].

Infeed tension, the most important zone for register, was set by adjusting draw-roll speed through a stepless speed-change gear. The ratio drifted and every cylinder change needed careful readjustment, so results depended heavily on operator skill [2].

B. Dancers, Powder Brakes and Analog Loops

The first real feedback came from the dancer roll: a roll on a pivoted arm loaded by dead weights, springs or an air cylinder. The dancer's load sets the tension, and its position tells the controller whether the brake is too strong or too weak. Magnetic particle (powder) brakes and clutches gave smooth, current-controlled torque at low speed and remain the cheapest unwind brake in use. Displacement-type tension detectors, whose bearings deflect microscopically under the web's wrap force, gave a direct tension signal.

Feedback replaced manual setting, and dancer-position feedback proved more effective than tension-signal feedback alone, especially during splicing and speed changes [2]. Rewinds gained diameter calculation, from a contact follower roller or from the ratio of line speed to spindle speed, which made taper tension possible [3].

C. DC Drives, PLCs and Automatic Register

DC motors on the infeed, outfeed and rewind, and powder brake plus DC motor combinations on the unwind, gave regenerative control over a wide speed and tension range: the powder brake carried high tension while the DC motor handled run-up and low tension [2]. Automatic register controllers read marks with photo-sensors and corrected register by moving compensator rollers that lengthen or shorten the web path between units. That correction itself disturbs tension, so tension and register loops had to be tuned together.

D. Electronic Line Shaft and Digital Control

The step change came when each print cylinder received its own motor, synchronised electronically instead of through a shaft. Valmet (Rotomec) stated that it took this step for gravure in 1996, and that almost all its gravure presses sold the following year were electronic line shaft models [10]. Each unit's cylinder phase can then be trimmed directly, and some makers now offer compensator-free register using feed-forward control [11]. AC vector motors replaced DC on the unwind and infeed, with less maintenance and a simpler explosion-proof design—important in solvent-ink areas [2].

IV. CURRENT METHODS

A modern gravure press measures tension with load cells and dancers, actuates it with AC vector or servo motors, and closes a separate loop in each zone under a supervisory controller that also runs register.

A. Sensing

Table IV compares the sensing options. Feeding back both dancer position and tension gives the best handling of splices and speed changes [2].

TABLE III
EVOLUTION OF TENSION CONTROL ON GRAVURE PRESSES

Era (approx.)	Tension method	Limitation that drove the next step
Up to 1950s–60s	Main motor and line shaft driving all units; friction band or shoe brakes on the unwind set by hand; rewind through a slip clutch; infeed draw set through a stepless speed-change gear	Tension depended on operator feel and drifted as roll diameter changed
1960s–1970s	Magnetic powder brakes and clutches; pneumatically loaded dancers; analog closed loops on dancer position or displacement-type tension detectors; diameter-compensated rewind	Analog drift, dancer inertia during splices, slow tuning
1980s–1990s	DC drives on infeed, outfeed and rewind; powder brake + DC motor unwinds; PLCs; automatic register control with compensator rollers	Mechanical line shaft still couples all units; register correction depends on compensators
Late 1990s–today	Electronic line shaft with a servo or AC vector motor per print unit; digital multi-zone loops; load cells plus dancers; recipe storage	Remaining limits: interaction between zones and thermal effects in dryers

TABLE IV
TENSION SENSING METHODS

Sensor	How it measures	Strengths	Watch-outs
Load-cell idler (strain gauge or micro-displacement)	Bearing force from web wrap; tension = force ÷ wrap factor	Direct, fast, no moving mass; suits thin and stiff webs	Calibrate with a hanging weight along the real web path; wrap angle must not change with roll diameter
Dancer roll with position sensor	Air-loaded arm; tension set by air pressure, position read by potentiometer or encoder	Stores web; absorbs splices, eccentricity and acceleration transients	Inertia and pivot friction; regulator accuracy sets tension accuracy
Dancer + load cell	Dancer absorbs transients, load cell trims true tension	Best during splices and speed changes	More tuning; two loops to coordinate
Sensorless (calculated)	Tension inferred from motor torque, diameter and losses	No hardware on the web path	Friction and inertia must be compensated; less accurate at low tension

B. Actuators

- *Unwind:* a magnetic powder brake on low-cost presses; a regenerative AC vector motor, often combined with a powder brake for high tension, on modern presses. Air brakes are not used on flexible packaging gravure presses because their low-torque behaviour is weaker than powder brakes [2].
- *Infeed and outfeed:* AC vector or servo motors driving nipped draw rolls, each speed-trimmed by its tension loop.
- *Print units:* one servo or AC vector motor per engraved cylinder on an electronic line shaft; impression rollers loaded pneumatically.
- *Rewind:* a centre-driven AC vector motor, sometimes with a lay-on roller for gap or surface winding to exclude air on films.

C. Control Architecture

1) *Speed master:* the line-speed reference, tied to the print units, sets every drive's base speed.
2) *Draw zones:* infeed and outfeed draw rolls hold tension by trimming their speed a fraction of a percent relative to the print units, using PI control on load-cell or dancer feedback.
3) *Torque zones:* unwind and rewind compute torque feedforward as set-point tension $\times$ radius, plus inertia compensation ($J\dot{\omega}$) and friction compensation, then trim with feedback.
4) *Register loop:* mark sensors after each unit measure longitudinal and lateral error, and the controller trims each cylinder's phase or moves a compensator. Because register trim changes web length, the register loop is tuned slower than the tension loops.
5) *Isolation:* nipped infeed and outfeed draw rolls isolate the print section, so rewind taper does not reach the last print unit [3].

D. Taper Tension at the Rewind

Rewind tension is reduced as the roll grows so that inner layers are not crushed and the roll does not telescope or block. With linear taper,

$$T(D) = T_0 \left[1 - \frac{A}{100} \cdot \frac{D - D_0}{D_{\max} - D_0} \right] \tag{2}$$

where T_0 is tension at the core diameter D_0 and A is the taper in percent. Taper of 30–50% is typical, rising to about 80% in some cases [4]. Modern controllers also allow hyperbolic or user-defined taper curves.

E. Supporting Features Now Standard

Recipe storage of tension and taper set points per substrate and job; stall tension and soft start; automatic splicing at full speed with dancers absorbing the transient; web preheating and chill rolls to stabilise film temperature, with paper over-dried to about 1–3% moisture [2]; and logging of tension, register error and speed together so the cause of a waste event can be traced.

V. Tension Zones from Unwind to Rewind

The nipped infeed and outfeed draw rolls split the press into three groups of zones, so that unwind and rewind disturbances do not reach the print units (Fig. 1).

- *Unwind zone:* the brake or motor holds the roll back against the infeed nip, cutting torque as the roll shrinks, absorbing eccentricity and riding through flying splices. A dancer is the most effective sensor here during splices.
- *Infeed zone:* the stability of infeed tension has the greatest influence on register precision, making the infeed the most important control unit on the press [2]. It is often paired with web preheating.
- *Print units and dryers:* between impression nips, tension is held by small speed differences (draw) between cylinders on the electronic line shaft. The web is softest inside the dryers, so thermoplastic films need the lowest tension and dryer temperature that still dries the ink.
- *Outfeed zone:* rewind tension strongly affects register of the last print unit; the outfeed roller blocks it and absorbs acceleration and splice transients [4].
- *Rewind zone:* builds a roll that must survive storage and the next process (lamination, slitting, pouch making), using taper (2).

VI. Guideline Tension Values

Set the unwind tension from the substrate's specific tension $\times$ thickness $\times$ width, then set every other zone as a ratio of it. The values are conservative starting points for solvent- or water-based gravure on flexible packaging presses, to be refined on each press (Section VII).

A. Basis of the Values

Film and foil values are taken from the widely used Dover Flexo Electronics running-tension chart in lb/in per mil [7], converted at 1 PLI/mil = 6.895 N/m per µm and expressed as a range: the lower bound suits heat-sensitive or thin gauges, the upper bound stiff, thick or wide webs. Paper and board values come from the same chart by basis weight and caliper. The ranges sit at roughly 3–12% of each film's yield stress, within Walker's guidance to run at about 10–20% of the damaging condition, or a 5:1 to 10:1 safety factor against yield [5]. Walker also notes that about 80% of webs run between 0.3 and 3.0 PLI (about 50–525 N/m) [5]. For further reference, TAPPI TIP 0200-01 gives conveyance and winding tension guidelines [8].

B. Effect of Web Width

Total tension rises in direct proportion to width; Table VII applies Tables V and VI to common gauges at three press widths. Width also matters beyond the total: wide webs (above about 1200 mm) carry more cross-web gauge variation and baggy lanes, so run them near the middle of the range and rely on spreader rolls rather than extra tension to remove wrinkles. Narrow webs on well-aligned, low-drag rollers can run closer to the upper limit [5].

TABLE V
Specific Unwind Tension for Films and Foil

Substrate	Thickness (µm)	N/m per µm (= MPa)	Notes
BOPET	10–36	3.5–5.2	Stiff, stable; easiest to register
Metallised BOPET/BOPP	as base	base −10 to 20%	Limit strain so metal layer does not crack
BOPP	15–40	1.4–2.4	Softens in dryers; lower end for pearlised, thin
CPP	20–50	1.0–1.7	Low modulus; watch neck-in
LDPE/LLDPE	30–100	0.5–1.2	Most tension-sensitive; low dryer temperature
HDPE	15–50	1.2–2.0	Stiffer than LDPE; prone to curl
BOPA (nylon)	12–25	1.5–2.5	Hygroscopic; keep wrapped
PVC/PETG shrink	30–50	0.6–1.2	Low tension and heat to avoid pre-shrink
Aluminium foil	20–40	3.5–6.0	Tears at nicked edges; avoid spikes

TABLE VI
Unwind Tension for Paper and Board

Substrate	Basis weight or caliper	Tension (N/m)
Paper	about 25 g/m^2	70
Paper	about 35 g/m^2	90
Paper	about 50 g/m^2	130
Paper	about 65 g/m^2	220
Paper	about 100 g/m^2	350
Paper	about 130 g/m^2	525
Board	200 µm (8 pt)	525
Board	300 µm (12 pt)	700
Board	380 µm (15 pt)	790
Board	500 µm (20 pt)	960
Board	760 µm (30 pt)	1400

TABLE VII
Total Unwind Tension, N (kgf in Brackets)

Substrate, gauge	600 mm	1000 mm	1300 mm
BOPET 12 µm	25–37 (2.6–3.8)	42–62 (4.3–6.4)	55–81 (5.6–8.3)
BOPET 23 µm	48–72 (4.9–7.3)	80–120 (8.2–12.2)	105–155 (10.7–15.8)
Met. BOPET 12 µm	21–32 (2.1–3.2)	35–53 (3.5–5.4)	45–69 (4.6–7.0)
BOPP 20 µm	17–29 (1.7–2.9)	28–48 (2.9–4.9)	36–62 (3.7–6.4)
BOPP 30 µm	25–43 (2.6–4.4)	42–72 (4.3–7.3)	55–94 (5.6–9.5)
CPP 30 µm	18–31 (1.8–3.1)	30–51 (3.1–5.2)	39–66 (4.0–6.8)
LDPE 50 µm	15–36 (1.5–3.7)	25–60 (2.5–6.1)	32–78 (3.3–8.0)
LDPE 80 µm	24–58 (2.4–5.9)	40–96 (4.1–9.8)	52–125 (5.3–12.7)
HDPE 25 µm	18–30 (1.8–3.1)	30–50 (3.1–5.1)	39–65 (4.0–6.6)
BOPA 15 µm	14–22 (1.4–2.3)	22–38 (2.3–3.8)	29–49 (3.0–5.0)
Shrink 40 µm	14–29 (1.5–2.9)	24–48 (2.4–4.9)	31–62 (3.2–6.4)
Al foil 25 µm	52–90 (5.4–9.2)	88–150 (8.9–15.3)	114–195 (11.6–19.9)
Paper 50 g/m^2	66–96 (6.7–9.8)	110–160 (11.2–16.3)	143–208 (14.6–21.2)
Board 300 µm	360–480 (37–49)	600–800 (61–82)	780–1040 (80–106)

C. Zone-Wise Ratios from Unwind to Rewind

Table VIII sets each zone relative to the unwind set point T_u. These ratios are the author's practical guideline compiled

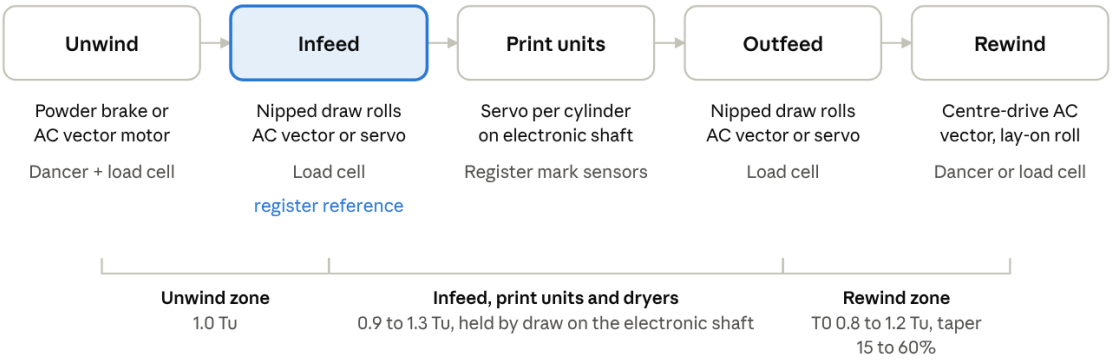


Fig. 1. Tension zones of a gravure press from unwind to rewind, with typical actuators and sensors at each station. Zone tensions are given as multiples of the unwind tension T_u (Table VIII). The infeed is the register reference.

from common gravure practice, not a published standard, and assume nipped infeed and outfeed draw rolls.

D. Worked Example

Consider a 1050 mm wide, 20 µm BOPP heat-seal film printed in eight colours and rewound from a 152 mm core to 800 mm diameter.

1) Unwind: 1.4–$2.4 \times 20 = 28$–48 N/m. Choosing the middle for a heat-sensitive film, about 36 N/m, gives $T_u = 36 \times 1.05 \approx 38$ N (3.9 kgf).
2) Infeed: $1.1 \times 38 \approx 42$ N.
3) Print units and dryers: 1.0–$1.1 \times 38 = 38$–42 N, with the lowest dryer temperature that dries the ink.
4) Outfeed: about $1.1 \times 38 \approx 42$ N.
5) Rewind: $T_0 = 38$ N at the core with 40% taper, falling to $38 \times 0.6 \approx 23$ N at 800 mm by (2).
6) Rewind torque: $38 \times 0.076 \approx 2.9$ N·m at the core and $23 \times 0.40 \approx 9.1$ N·m at full diameter; the motor and gearbox must cover this range with margin.

If the printed repeat comes out short, tension is too high and should be lowered; if long, it should be raised slightly [2].

VII. Commissioning, Tuning and Troubleshooting

Most tension problems trace back to an uncalibrated sensor, a misaligned roller or a loop tuned faster than the register loop, not to a wrong set point.

A. Commissioning Checklist

1) Calibrate every load cell with a known weight hung over the real web path and wrap angle; record zero and span.
2) Where possible, verify by hand: a spring balance pulled on the web until the unwind turns reads total tension directly [6].
3) Check roller parallelism and level along the full path; misalignment causes wrinkles that no tension setting cures.
4) Check dancer regulator accuracy and pivot friction; the dancer should float mid-stroke at steady speed.
5) Set inertia and friction compensation on unwind and rewind with an empty run-up before tuning feedback gains.
6) Tune zones from the infeed outward: infeed, outfeed, unwind, rewind; tune the register loop last and slower.
7) Save tested set points as recipes per substrate, gauge and width.

B. Symptom Guide

Table IX links common defects to their likely tension causes.

VIII. Conclusion

Tension control in gravure has moved from operator feel to engineered, measured control, and the register gains of the modern press depend on it. The mechanical press tied every unit to one shaft and left tension to brakes, clutches and speed-change gears that drifted with roll diameter. Dancers, powder brakes and analog loops brought feedback; DC drives and PLCs brought isolation between zones; the electronic line shaft and AC vector drives brought per-unit control, recipes and compensator-free register.

The physics has not changed: tension sets strain, and strain sets register. Choosing the right tension therefore starts from the substrate's modulus, thickness and width. The guideline tables give starting values for common gravure substrates and zone ratios from unwind to rewind. Each press, ink system and dryer setting will move the best value, so the tables should be refined with calibrated sensors and logged register data, and proven set points saved as recipes.

Acknowledgment

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TABLE VIII
ZONE TENSION AS A MULTIPLE OF UNWIND TENSION T_u

Zone	Stiff: BOPET, foil, paper, board	Medium: BOPP, HDPE, BOPA	Stretchy/heat-sensitive: LDPE, CPP, shrink	Purpose
Unwind	1.0	1.0	1.0	Hold back the roll; isolate eccentricity and splices
Infeed	1.1–1.3	1.0–1.2	1.0–1.1	Register reference; the most stable zone
Print units, dryers	1.1–1.3	1.0–1.2	0.9–1.0	Hold register without stretching softened film
Outfeed	1.1–1.3	1.0–1.2	0.9–1.1	Isolate the print section from rewind taper
Rewind at core (T_0)	1.0–1.2	0.9–1.1	0.8–1.0	Start firm so the core does not slip or star
Rewind taper (A)	15–30%	30–50%	35–60%	30–50% typical, up to about 80% [4]

TABLE IX
TROUBLESHOOTING TENSION-RELATED DEFECTS

Symptom	Likely tension cause	First actions
Register drifts during acceleration or deceleration	Inertia compensation missing or wrong; infeed loop too slow	Set unwind and rewind inertia compensation; check infeed load-cell response
Once-per-revolution register wander	Eccentric unwind roll; dancer too stiff	Check roll runout; let the dancer absorb it; lower unwind gain
Register error at a splice	Dancer bottoms out; splice tension mismatch	Pre-match new-roll speed; check dancer stroke and loading
Printed repeat short	Tension too high for film modulus and temperature	Lower print-zone tension and dryer temperature
Printed repeat long; web flutters	Tension too low	Raise tension within the Table V range
Machine-direction wrinkles or creases	Uneven cross-web tension, baggy web, misaligned rollers	Align rollers; add spreader rolls; do not raise tension above range
Neck-in, curl or gauge bands on PE, CPP, shrink film	Tension too high in hot dryers	Use stretchy-web ratios (Table VIII); reduce dryer temperature and air velocity
Cracked metal layer on metallised film	Strain above the coating's limit	Use metallised values (base film -10 to 20%)
Telescoping or loose rewound roll	Rewind tension too low at core, or too much taper	Raise T_0; reduce taper; check lay-on roller
Starring, crushed core, blocking	Rewind too tight; too little taper	Increase taper; reduce T_0
Last colour only out of register	Rewind disturbance reaching the print section	Close and control the outfeed nip; check outfeed loop

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